Privacy-Preserving Computer Vision for Airport Biometrics

Jason Stout
NC State University
Raleigh, NC, USA
jcstout3@ncsu.edu

Mounir Arayssi
NC State University
Raleigh, NC, USA
marayss@ncsu.edu

Your Name
NC State University
Raleigh, NC, USA
your_email@example.com

Your Name
NC State University
Raleigh, NC, USA
your_email@example.com

Abstract

1 Introduction

Facial recognition technology is increasingly being deployed in airports to streamline identity verification, improve security, and reduce passenger processing time. These systems, used by agencies such as the U.S. Customs and Border Patrol (CBP) and similar organizations worldwide, enable automated matching of travelers' biometric data against government databases. Despite the efficiency and security benefits of these systems, they also introduce many significant concerns regarding privacy and data protection.

Biometric identifiers such as facial scans and fingerprints are inherently sensitive and immutable. Unlike passwords, compromised biometric data cannot be changed, making misuse or breaches particularly harmful. In addition, the large scale collection and storage of biometric data in centralized systems raises concerns over surveillance, function creep, erosion of privacy, and the security of said centralized systems.

This project investigates the privacy implications of facial recognition systems in airport environments by analyzing existing legal frameworks, identifying key privacy risks, and evaluating current technical protections. Building on this analysis, we propose a privacy-preserving system framework that integrates computer vision with privacy-by-design principles. To demonstrate how these protections can be applied in practice, we implement a proof-of-concept system using representative airport scenarios. Although these scenarios focus on general detection and monitoring, the same framework can be extended to identify verification systems such as biometric entry and exit processes.

2 Related Work

2.1 Legal Frameworks for Biometric Data

Regulation of biometric data varies significantly across regions. The European Union provides one of the most comprehensive frameworks through the General Data Protection Regulation (GDPR), which classifies biometric data as a special category requiring explicit consent or a strict legal basis for processing. Under GDPR, organizations must follow strict requirements such as data minimization, purpose limitation, and the use of data protection impact assessments (DPIAs) when handling sensitive biometric information.

Additional regulations, such as the Entry/Exit System (EES), mandate the collection of biometric data for non-EU travelers while enforcing safeguards such as data retention periods and accountability measures. However, the United States relies on a combination of federal and state-level laws, such as the Illinois Biometric Information Privacy act (BIPA), resulting in a more fragmented approach to regulation.

While these existing legal frameworks establish important principle for protecting biometric data, they often lack clear transparent guidelines on how such protections should be implemented in real-world airport systems. Many deployments continue to rely on centralized data storage, limited transparency, and opt-out mechanisms that make it not clearly explained to

travelers. This highlights a gap between regulatory principles and actual system implementation.

2.2 Technical Frameworks and Standards

Technical standards emphasize privacy-preserving approaches to biometric systems. ISO/IEC 24745 outlines key principles such as irreversibility, unlinkability, and secure storage of biometric templates. Instead of storing raw images, systems are encouraged to use one-way transformed templates that cannot reconstruct the original biometric data.

Additional standards, including ISO/IEC 30107, address presentation attack detection, while frameworks from organizations such as NIST and IATA promote privacy-by-design principles. Emerging approaches, such as self-sovereign identity and verifiable credentials, aim to give users greater control over their biometric data by enabling decentralized identity management.

While these standards provide clear guidelines for building secure systems, they are not always consistently used in real-world airport deployments. Many systems still rely on centralized storage and do not fully apply these privacy principles, highlighting the need for practical system designs that integrate these standard directly into operational pipelines.

3 Methodology

Our approach focused on analyzing both legal and technical aspects of facial recognition systems in airport environments. First, we conducted a comparative review of international legal frameworks, examining how different regions regulate the collection, storage, and use of biometric data. This included evaluating regulatory requirements, enforcement mechanisms, and user protections.

Next, we analyzed technical standards and best practices proposed by organizations such as ISO, NIST, and IATA. This involved reviewing recommended approaches for secure storage, encryption, biometric template protection, and privacy-preserving system design.

We then identified key privacy risks associated with current deployments by combining findings from academic literature, industry reports, and advocacy organizations. These risks were then evaluated alongside existing mitigation strategies, including both legal safeguards and technical solutions.

Finally, our team applied these insights to inform the design of an alternative conceptual system aimed at balancing airport security needs with enhanced privacy protections.

3.1 System Design

Based on the identified risks and existing frameworks, we designed a privacy-preserving system composed of three main components: a perception layer, a decision layer, and a privacy layer. The perception layer uses computer vision techniques to detect relevant objects in airport environments, such as passengers and luggage. In our implementation, this was achieved using a pre-trained object detection model to identify specific classes including people, backpacks, handbags, and suitcases.

The decision layer classifies detected events into categories such as normal activity, operational issues, or potential security threats. This classification is based on simple rules derived from detected objects, such as identifying high congestion when multiple individuals are present or flagging potential risks when luggage is detected without nearby individuals.

The privacy layer enforces privacy-preserving mechanisms throughout the system. Instead of storing raw biometric data, the system converts detected biometric information into secure templates using one-way transformations. In addition, faces are blurred before any further processing occurs, ensuring that identifiable facial data is not retained. In this prototype, face blurring is used to demonstrate post-processing privacy protection. In real biometric entry or exit systems, facial data would be used temporarily for identity matching before being transformed into secure templates or deleted, rather than being permanently stored.

3.2 Prototype Implementation

To demonstrate the feasibility of the proposed system, we developed a proof-of-concept implementation using real airport scenarios. Sample images were used to simulate real-world environments, and a pre-trained YOLO-based object detection model was applied to identify relevant objects such as individuals and luggage.

To enforce data minimization, the system filters the detection results to include only relevant classes, discarding all other detected objects. Each image is processed through a pipeline where faces are first blurred,

object detection is performed, and then detected regions associated with individuals are converted into hashed biometric templates instead of being stored as raw image data.

The system classifies scenarios into categories such as normal passenger activity, high congestion, and potential security risks such as unattended baggage. For example, a scene containing unattended luggage is flagged as a potential security alert, while a large number of detected individuals indicates congestion. These classifications show how the system can support both operational awareness and security monitoring while maintaining privacy protections. These scenarios were used to evaluate how these mechanisms affect system behavior and to highlight tradeoffs between security functionality and data protection.

3.3 Scope and Application

While the implementation focuses on general airport monitoring scenarios, it serves as a proof-of-concept for demonstrating how privacy-preserving techniques can be applied to biometric systems in airports. The use of generalized scenarios allows for testing of privacy mechanisms across different conditions without relying on sensitive real-world biometric datasets.

The same framework can be extended to identity verification systems, such as facial recognition-based entry and exit processes, where privacy risks are often more significant due to the direct use of biometric identity matching. This approach allows for evaluation of privacy protections across multiple stages of airport operations rather than limiting the analysis to a single use case.

4 Findings

4.1 System Output Interpretation

The output of the system is visualized using bounding boxes over detected objects within each image. Each bounding box is labeled with the object class (e.g., person, suitcase, backpack) along with a confidence score, which represents the model's estimated probability of a correct detection. These detections are produced by the perception layer and serve as the input for the decision layer, which classifies each scenario.

4.2 Normal Activity Scenario

In this scenario, the system detects a passenger and their luggage and classifies the scene as normal activity. The perception layer identifies objects such as a person and a suitcase, and the decision layer uses this information to determine the event type. During processing, faces are blurred and no raw biometric data is stored.

This demonstrates that standard monitoring tasks can still be performed while applying privacy-preserving techniques such as data minimization and template-based representations. The system maintains functionality while limiting the exposure of sensitive data.

Takeaway: Routine monitoring can be performed without storing identifiable biometric data.



Figure 1: Normal passenger scenario with object detection and privacy-preserving processing.

4.3 High Congestion Scenario

In this scenario, the system detects many individuals and classifies the scene as high congestion. Even with a large number of detections, the system continues to filter out unnecessary data and does not store sensitive biometric information. The classification is based on the number of people detected in the scene.

Takeaway: Privacy protections can scale to crowded, real-world conditions.

4.4 Security Alert Scenario

In this scenario, the system detects unattended baggage and flags it as a potential security risk. This classification is based on the presence of luggage without any nearby individuals. The system is able to identify this situation without relying on stored biometric data.

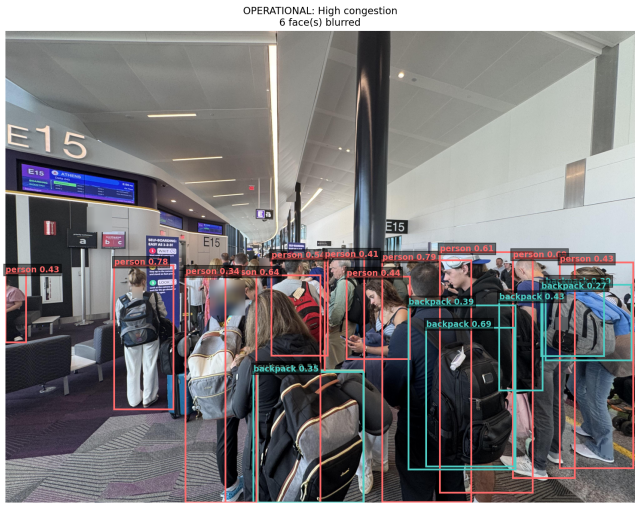


Figure 2: High congestion scenario with multiple detected individuals.

Takeaway: Security alerts can be generated without long-term biometric data storage.

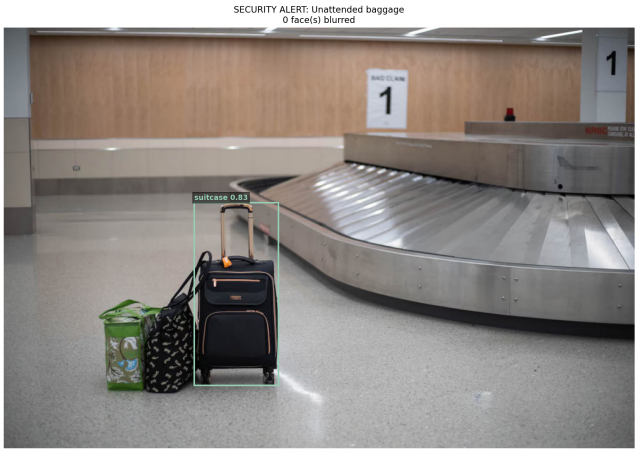


Figure 3: Unattended baggage detected and classified as a security alert.

4.5 Overall Observations

Across all scenarios, the system demonstrates that privacy-preserving techniques can be integrated into computer vision pipelines without completely limiting functionality. The system is able to detect objects, classify events, and generate alerts while avoiding the storage of raw biometric data. However, these results also highlight that privacy protections require tradeoffs, particularly when tasks such as identity verification are needed.

4.6 Tradeoffs and Limitations

The proposed system illustrates that privacy-preserving techniques can coexist with airport monitoring functions, but several tradeoffs and limitations remain.

The most significant tradeoff is between privacy and functionality. Mechanisms such as face blurring and one-way template conversion protect identifiable data, but they also restrict the system's ability to perform identity matching, watchlist screening, or longitudinal tracking. In a deployed biometric entry or exit system, these protections would need to be applied selectively rather than uniformly, since some operations require temporary access to identifiable features before transformation or deletion.

The decision layer also relies on simplified, rule-based classification. Events such as congestion or unattended baggage are inferred from object counts and spatial relationships rather than learned behavioral models. This keeps the prototype interpretable and auditable, but it limits robustness in complex scenes where context, motion, or temporal patterns would be needed to distinguish genuine threats from benign activity.

Evaluation was conducted on a small set of representative sample images rather than real airport footage. As a result, the system was not benchmarked using quantitative metrics such as detection accuracy, false positive rate, latency, or throughput under load. We also did not assess performance across demographic groups, which is a known source of bias in facial recognition systems and a critical concern for any production deployment.

Finally, the prototype operates outside of real infrastructure. It does not account for camera placement, network constraints, integration with existing CBP or airline systems, or the operational requirements of high-volume terminals. The choice between centralized and local processing illustrates a related tradeoff: on-device matching reduces breach exposure and aligns with data minimization, but it increases system complexity, hardware cost, and the difficulty of model updates and auditing.

5 Future Directions

Several directions would extend this work toward a deployable system. The most immediate next step is implementing a real-time detection pipeline tested on representative airport video data rather than static sample images. This would enable quantitative evaluation across standard metrics, including detection accuracy, latency, and throughput, and would allow the privacy mechanisms to be measured for their performance overhead rather than only their conceptual coverage.

A second direction is fairness and bias evaluation. Facial recognition systems have well-documented disparities in accuracy across demographic groups, and any privacy-preserving variant must be evaluated under the same lens. Future work should test the system across diverse populations and report per-group error rates alongside aggregate performance.

A third direction is strengthening the privacy layer with more advanced cryptographic techniques. Hashed templates provide irreversibility but offer limited protection during matching itself. Approaches such as homomorphic encryption, secure multi-party computation, and trusted execution environments would allow biometric comparisons to be performed without ever exposing the underlying template in plaintext. On-device or edge-based matching using these techniques could further reduce reliance on centralized storage.

A fourth direction is user-facing transparency and control. Integration with self-sovereign identity systems and verifiable credentials would let travelers carry their own cryptographic identity proofs and present them selectively at checkpoints, shifting the trust model away from centralized government databases. This would also support clearer consent flows and opt-out mechanisms, addressing the gap between regulatory principles and current deployment practices identified earlier in the paper.

Finally, future work should examine the operational and policy layer surrounding these systems. Technical safeguards are most effective when paired with enforceable retention policies, independent audits, and clear escalation procedures for human review. Evaluating how privacy-preserving computer vision integrates with these governance structures, rather than treating it as a purely technical problem, is essential for translating proof-of-concept systems into responsible real-world deployments.

6 Team Contribution

- Jason Stout: Conducted research on legal frameworks, technical standards, privacy risks, and mitigation strategies related to biometric systems.
- Mounir Arayssi: Researched alternative approaches to traditional biometric systems and contributed to the design and testing of the proposed system.
- Jude El Masri: Researched alternative approaches to traditional biometric systems and combined team research for finalized approach to new system
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7 Conclusion

References